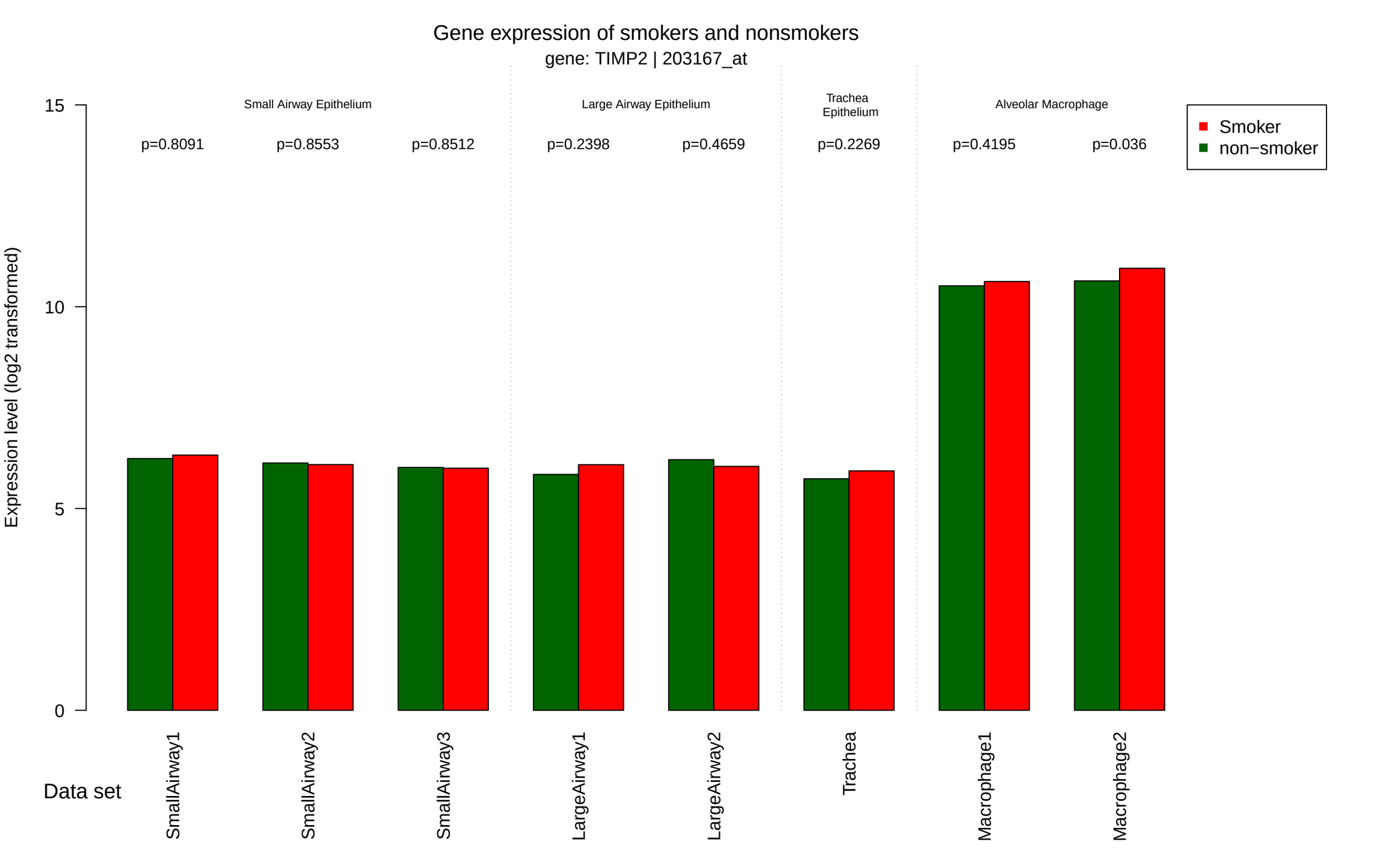
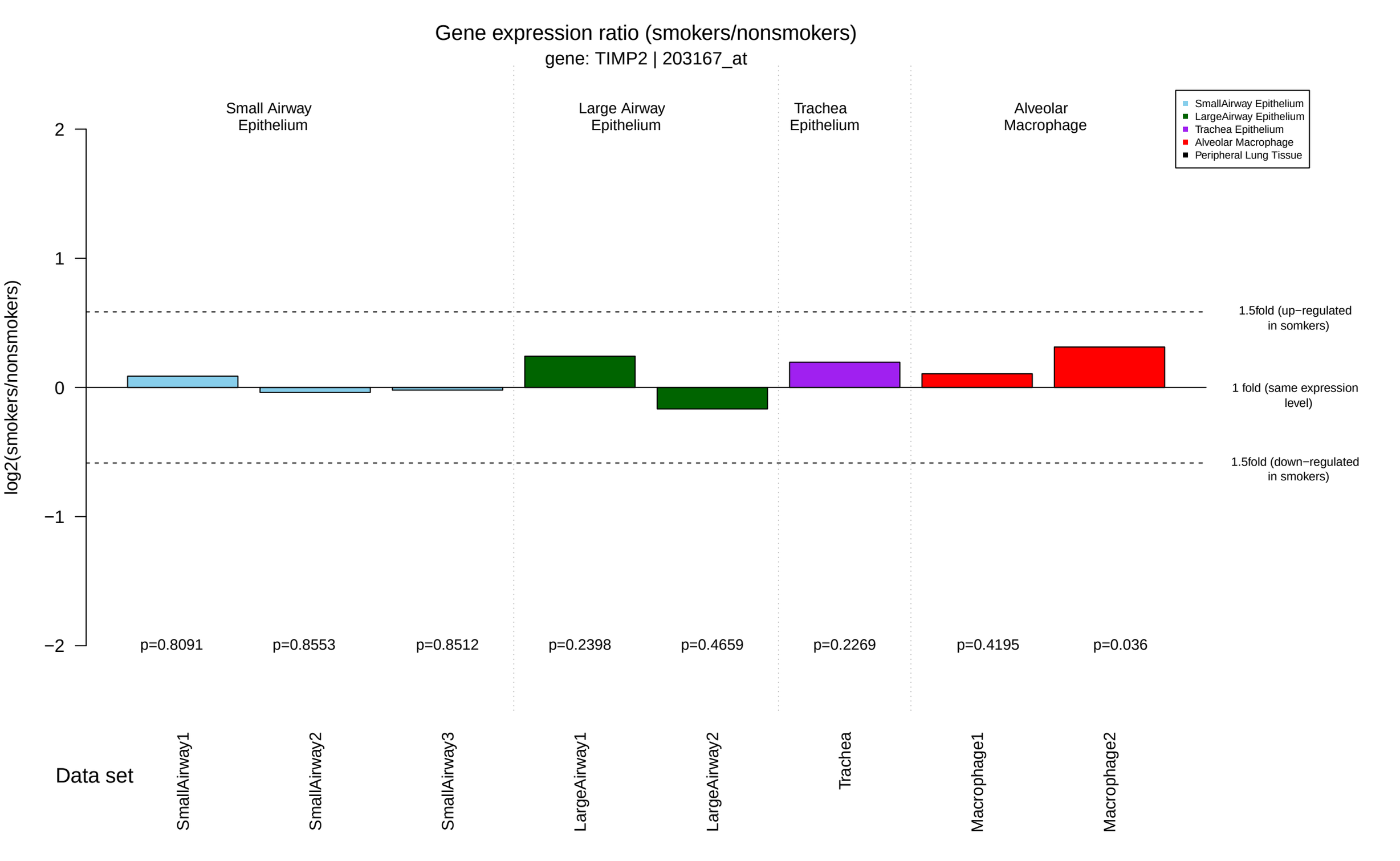
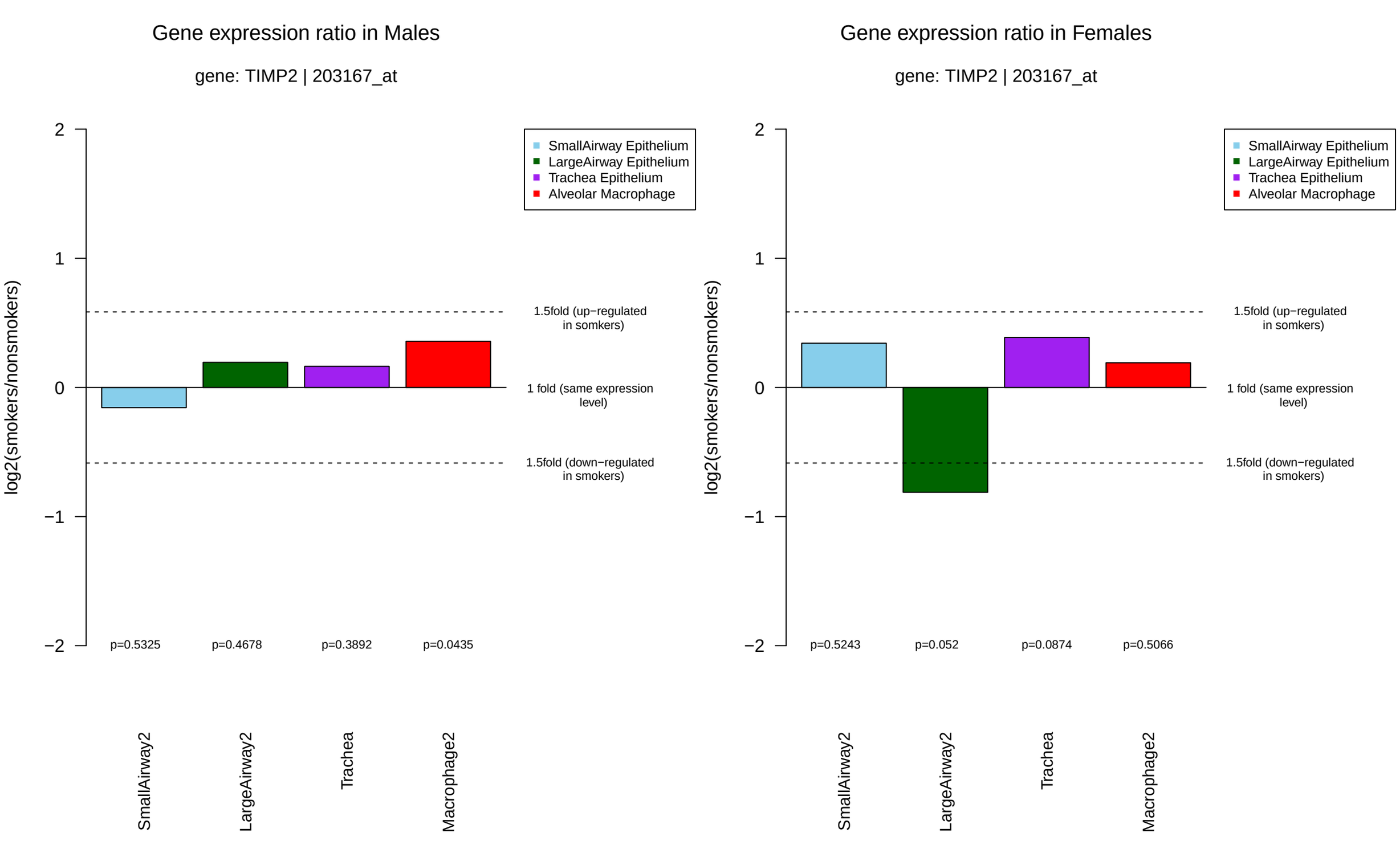
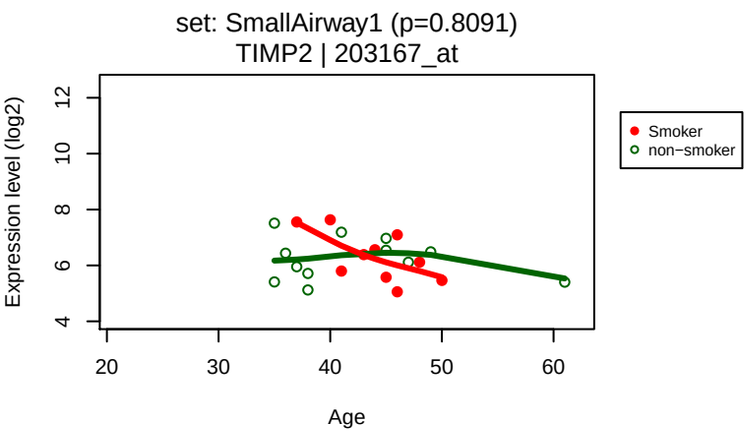
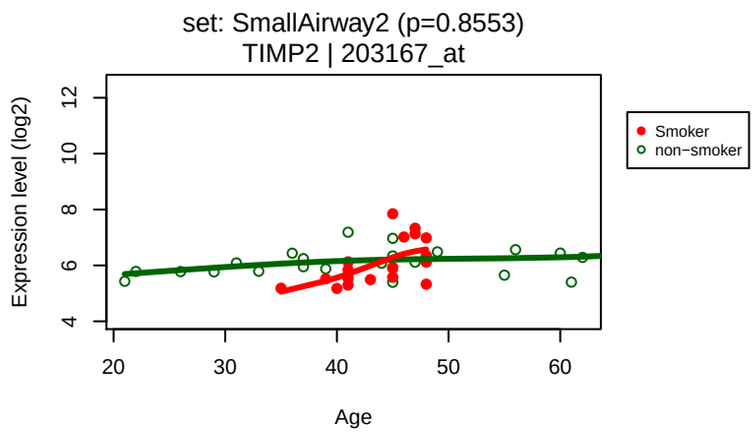
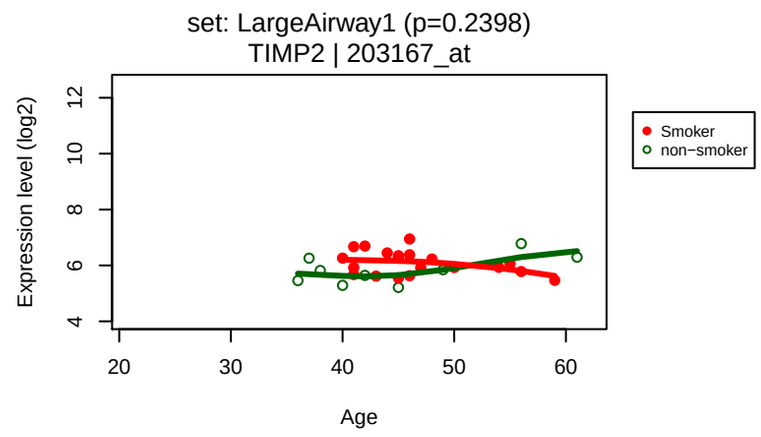

Gene expression of smokers and nonsmokers
gene: TIMP2 | 203167_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8091
p=0.8553
p=0.8512
p=0.2398
p=0.4659
p=0.2269
p=0.4195
p=0.036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

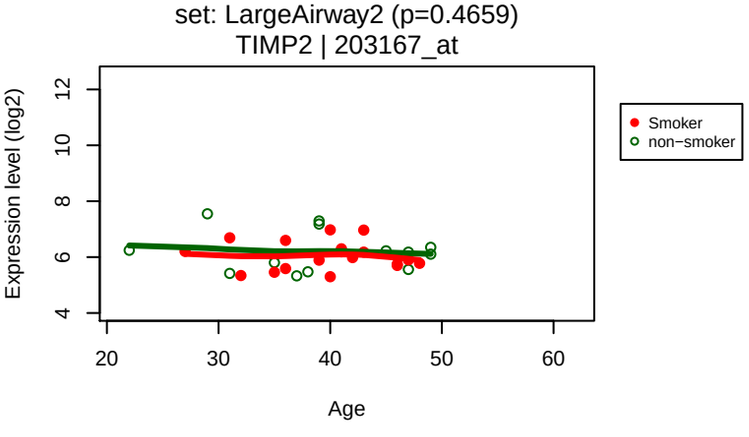

Gene expression ratio (smokers/nonsmokers)
gene: TIMP2 | 203167_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8091
p=0.8553
p=0.8512
p=0.2398
p=0.4659
p=0.2269
p=0.4195
p=0.036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

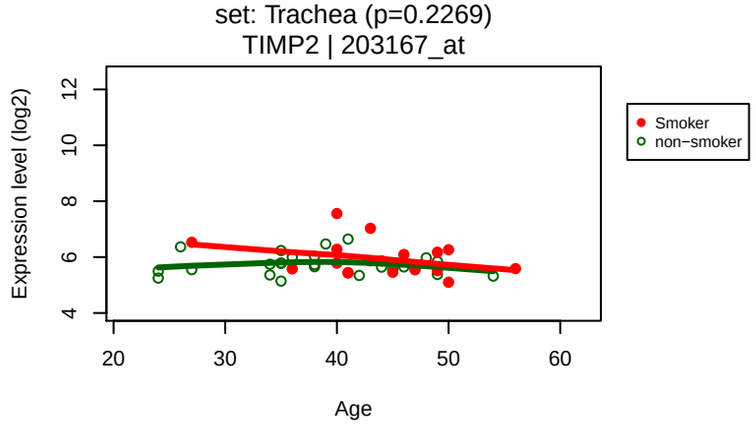

Gene expression ratio in Males
gene: TIMP2 | 203167_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5325
p=0.4678
p=0.3892
p=0.0435
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP2 | 203167_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5243
p=0.052
p=0.0874
p=0.5066
SmallAirway2
LargeAirway2
Trachea
Macrophage2

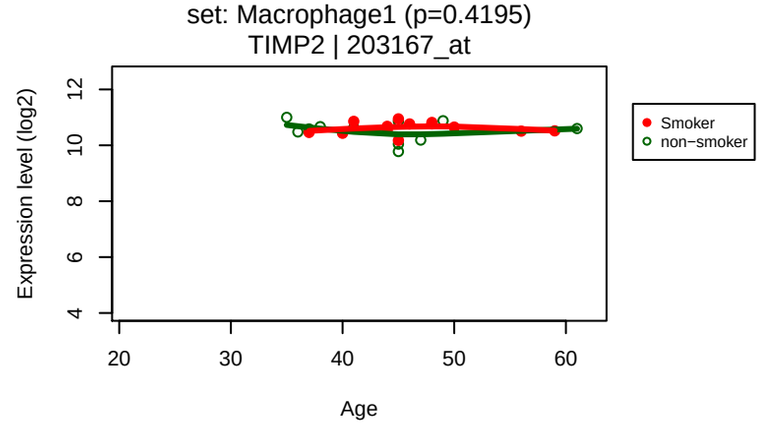

set: SmallAirway1 (p=0.8091)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker

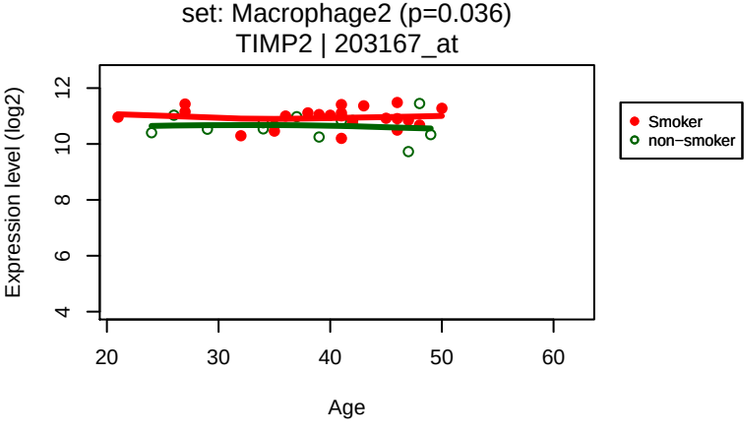

set: SmallAirway2 (p=0.8553)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker

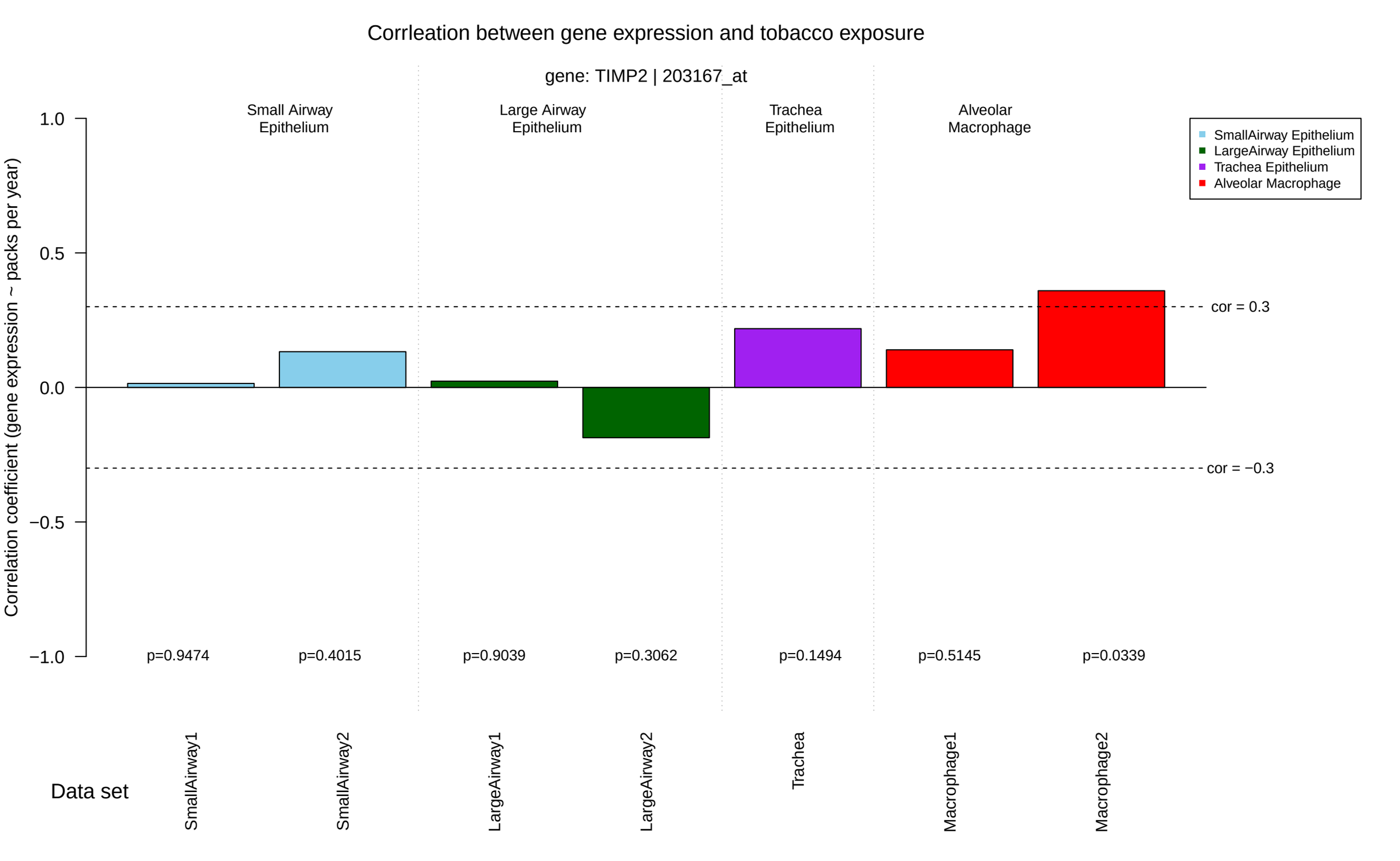

set: LargeAirway1 (p=0.2398)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4659)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2269)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4195)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.036)
TIMP2 | 203167_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP2 | 203167_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9474
p=0.4015
p=0.9039
p=0.3062
p=0.1494
p=0.5145
p=0.0339
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2